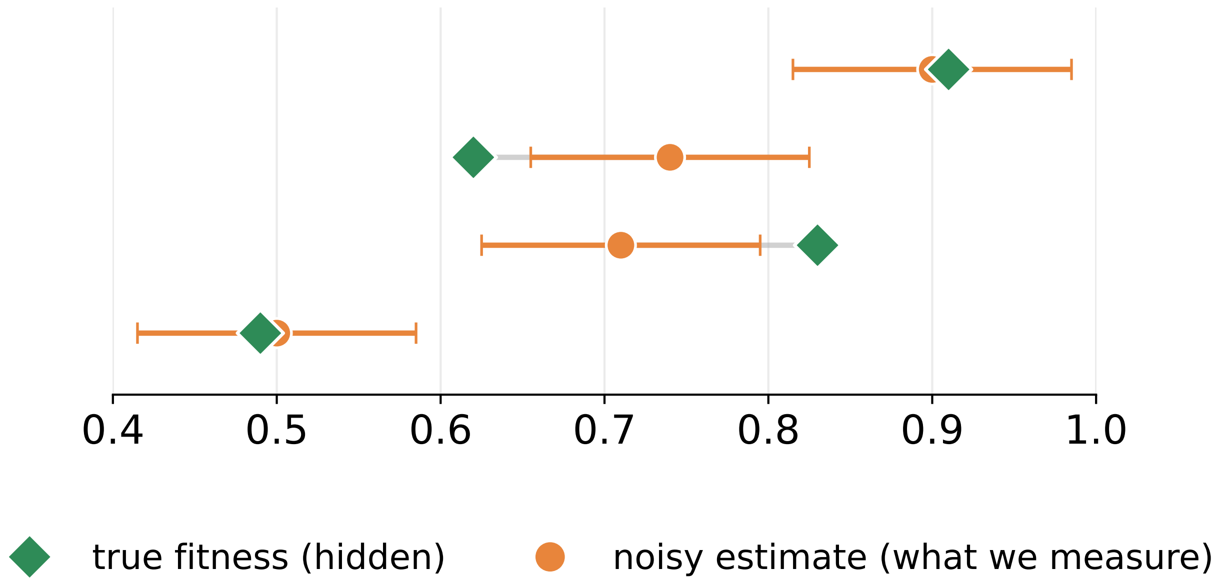
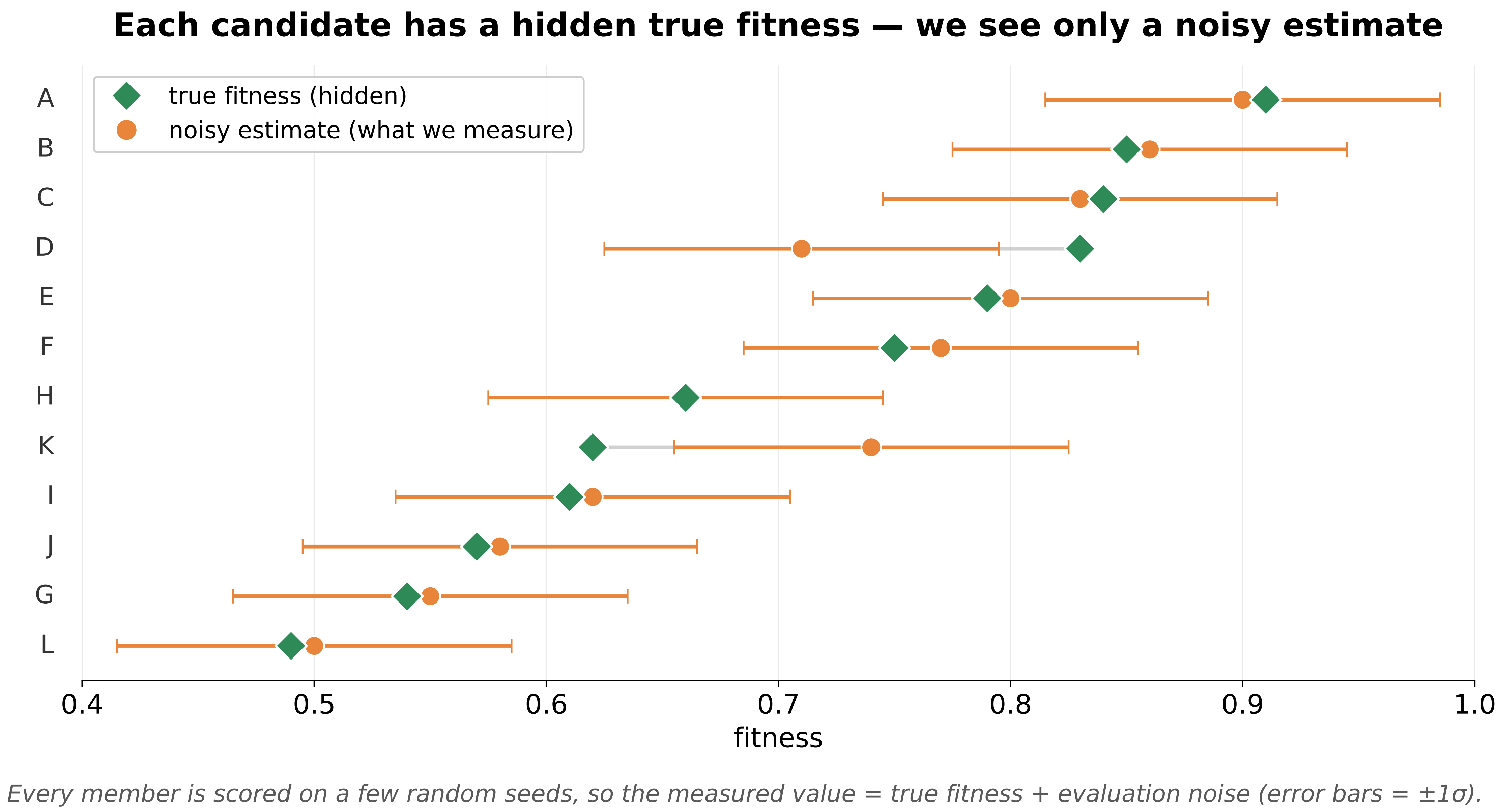
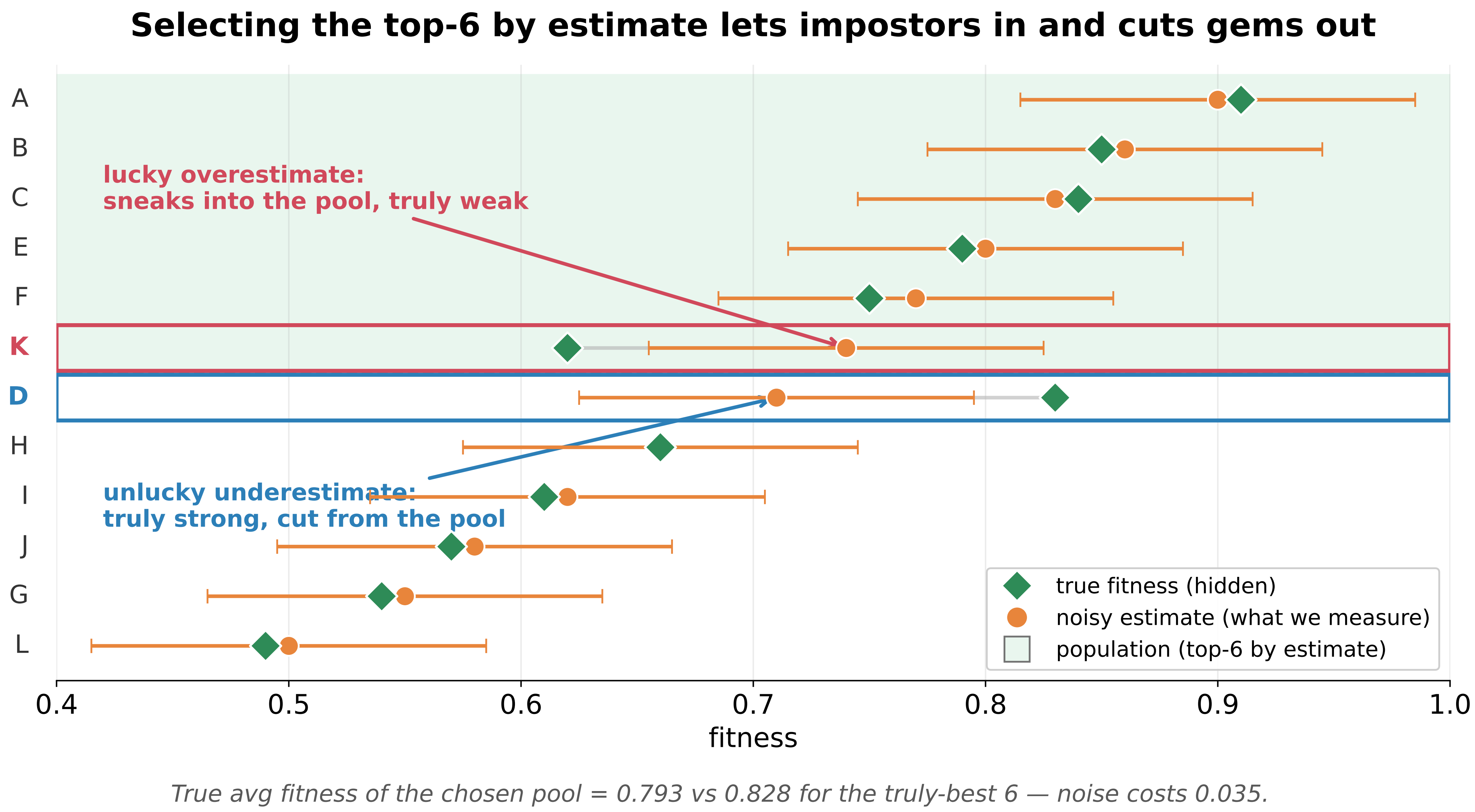
Why reevaluate?

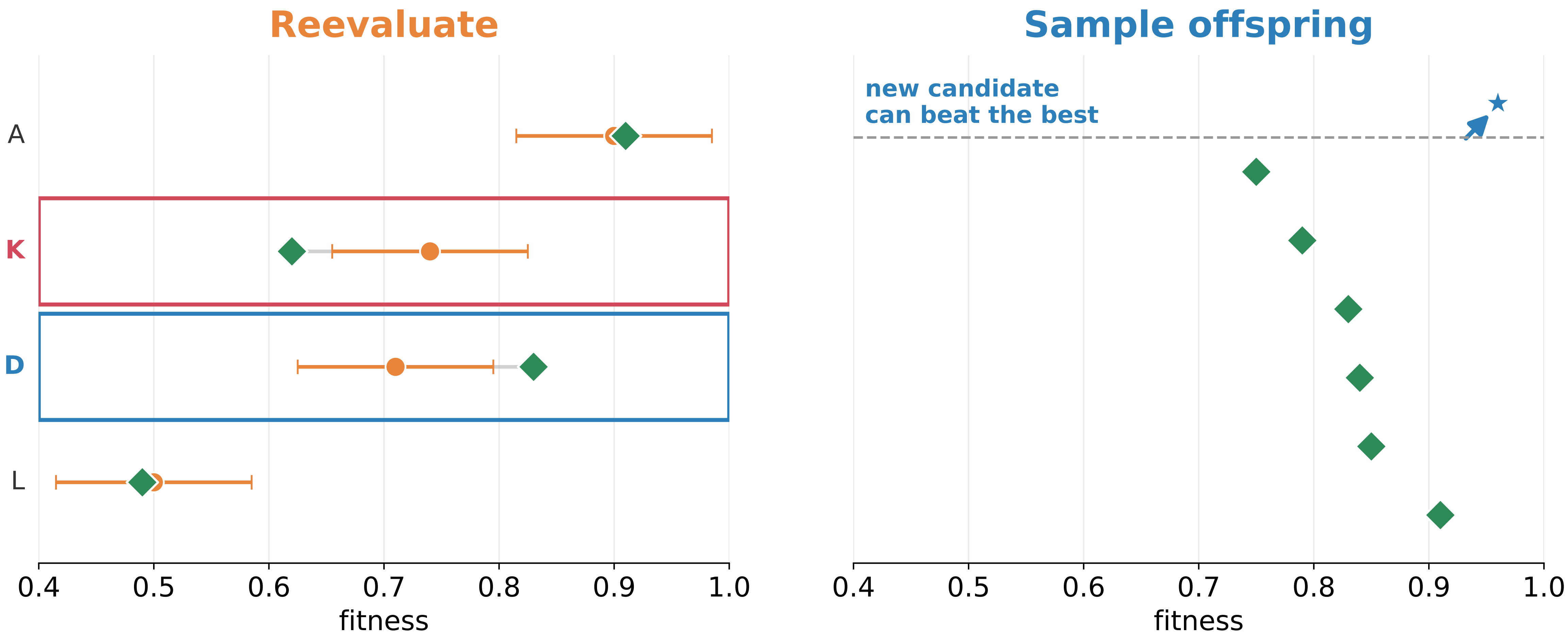
Noise, selection, and the value of a second look




Each candidate has a hidden true fitness — we see only a noisy estimate
true fitness (hidden)
noisy estimate (what we measure)
A
B
C
D
E
F
H
K
I
J
G
L
0.4
0.5
0.6
0.7
0.8
0.9
1.0
fitness
Every member is scored on a few random seeds, so the measured value = true fitness + evaluation noise (error bars = ±1σ).


Selecting the top-6 by estimate lets impostors in and cuts gems out
A
B
C
E
F
K
D
H
I
J
G
L
lucky overestimate:
sneaks into the pool, truly weak
unlucky underestimate:
truly strong, cut from the pool
true fitness (hidden)
noisy estimate (what we measure)
population (top-6 by estimate)
0.4
0.5
0.6
0.7
0.8
0.9
1.0
fitness
True avg fitness of the chosen pool = 0.793 vs 0.828 for the truly-best 6 — noise costs 0.035.

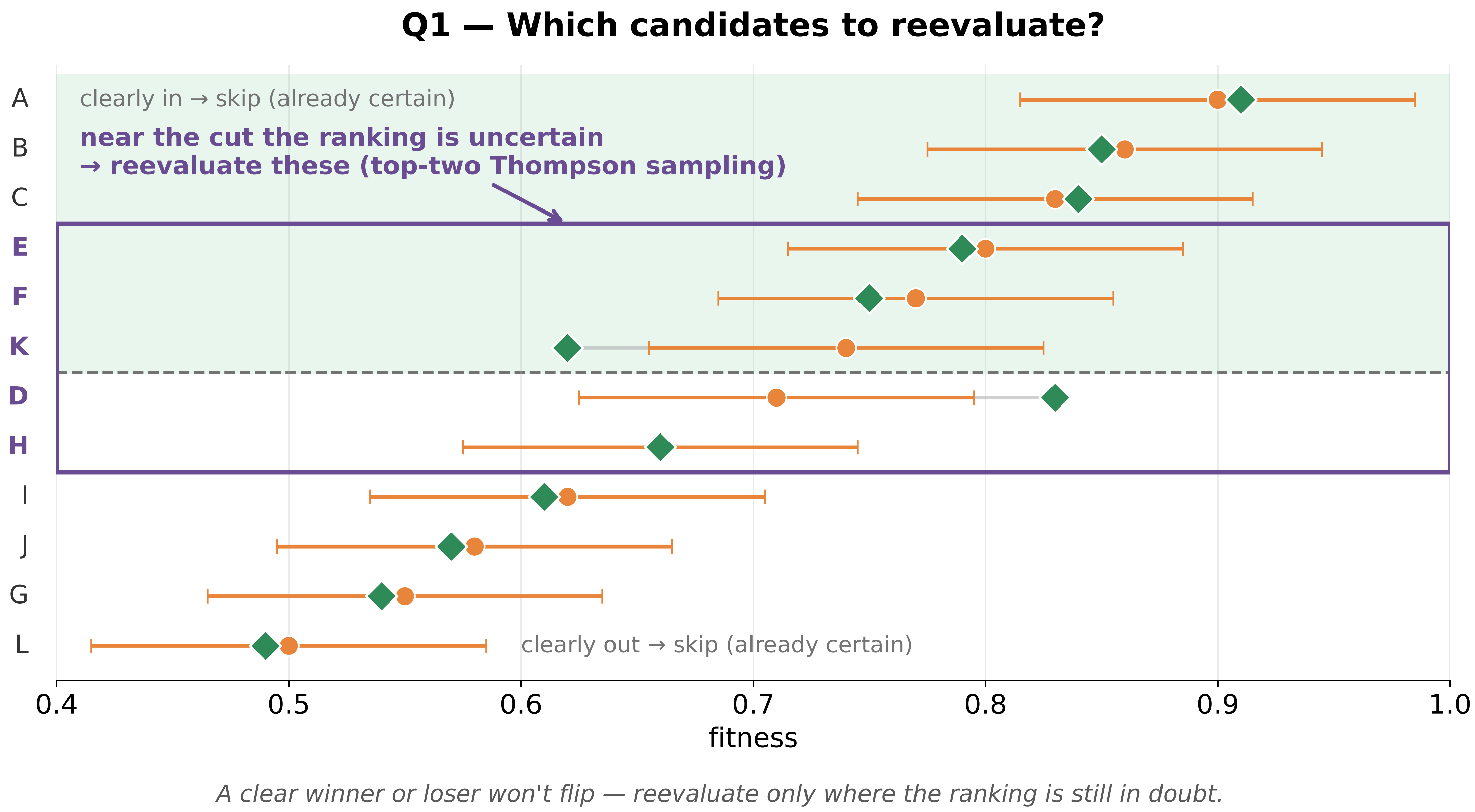
Two levers to raise the next generation's parent fitness

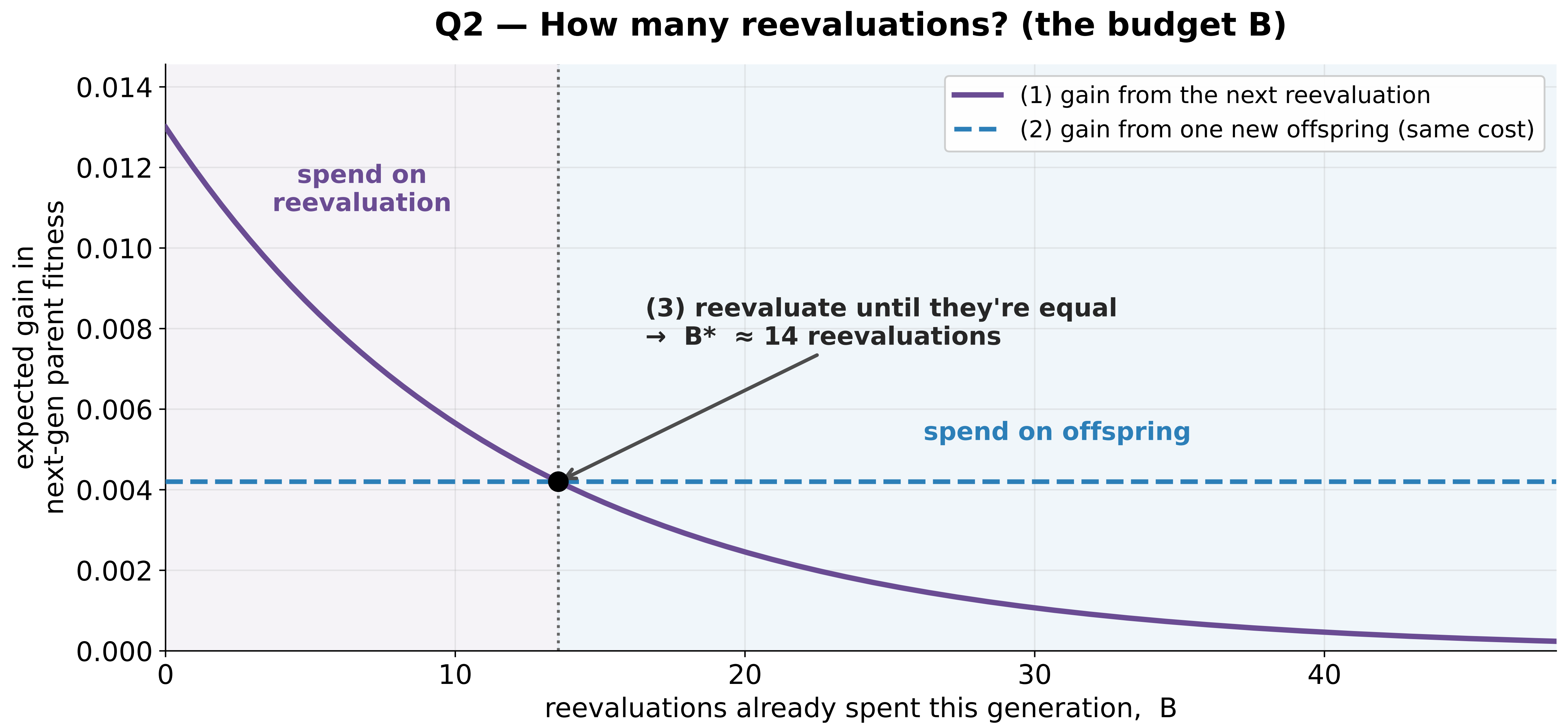


Sharpen estimates of existing members
→ better accuracy, fix who is in the pool

Introduce new members near good parents
→ raises the ceiling of achievable fitness

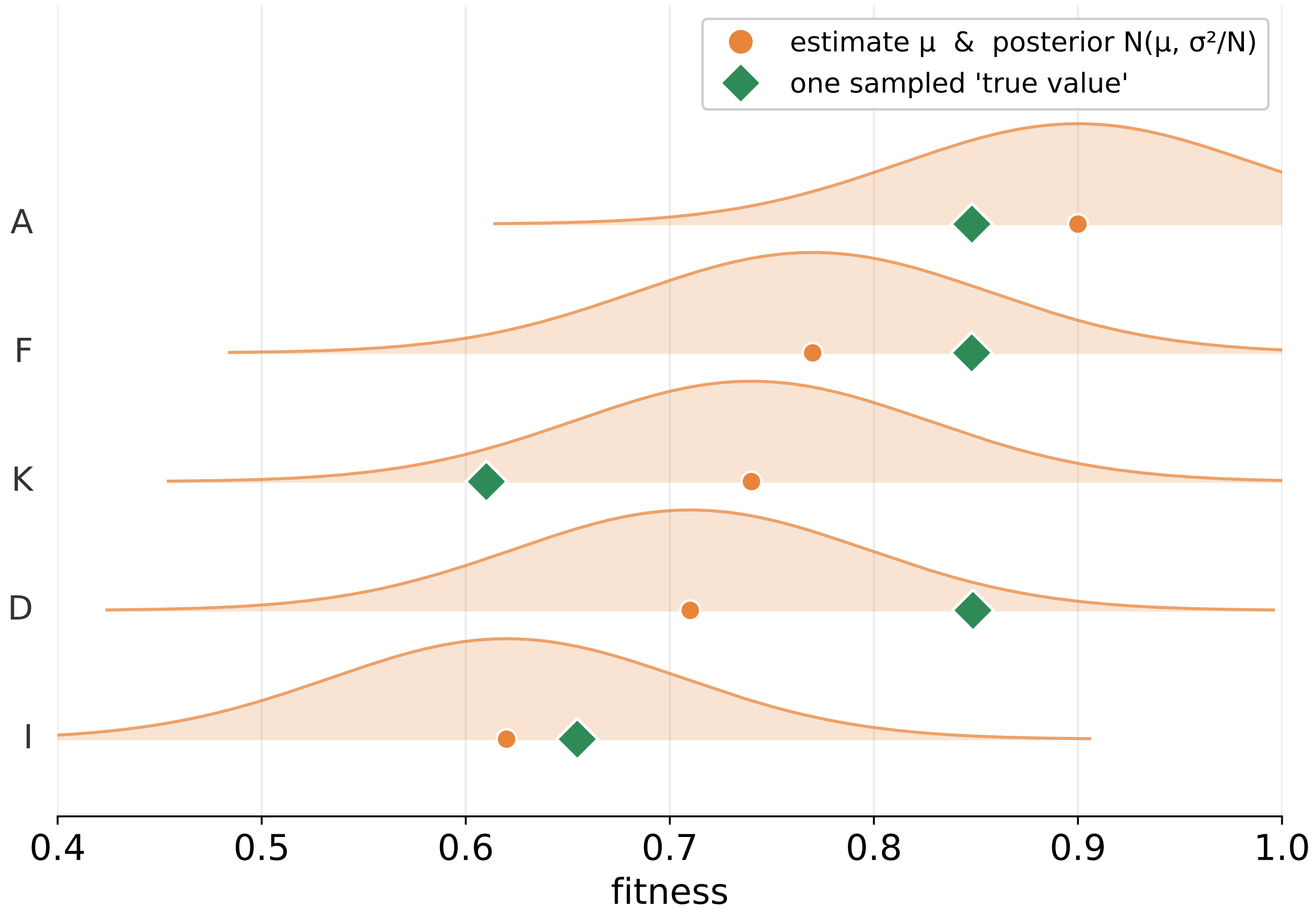
Offspring and reevaluations cost the same seeds; smart reeval spends them where the marginal gain is larger (the B point).*


Q1 — Which candidates to reevaluate?
clearly in → skip (already certain)
near the cut the ranking is uncertain
→ reevaluate these (top-two Thompson sampling)
clearly out → skip (already certain)
A
B
C
E
F
K
D
H
I
J
G
L
0.4
0.5
0.6
0.7
0.8
0.9
1.0
fitness
A clear winner or loser won't flip — reevaluate only where the ranking is still in doubt.



Goal: maximize next-gen parent fitness. Each reeval buys less than the last — keep going only while it beats one offspring.

Reevaluation EI — step 1: imagine the true values



Each arm's belief is a posterior $\theta \sim N(\mu, \sigma^2/N)$.

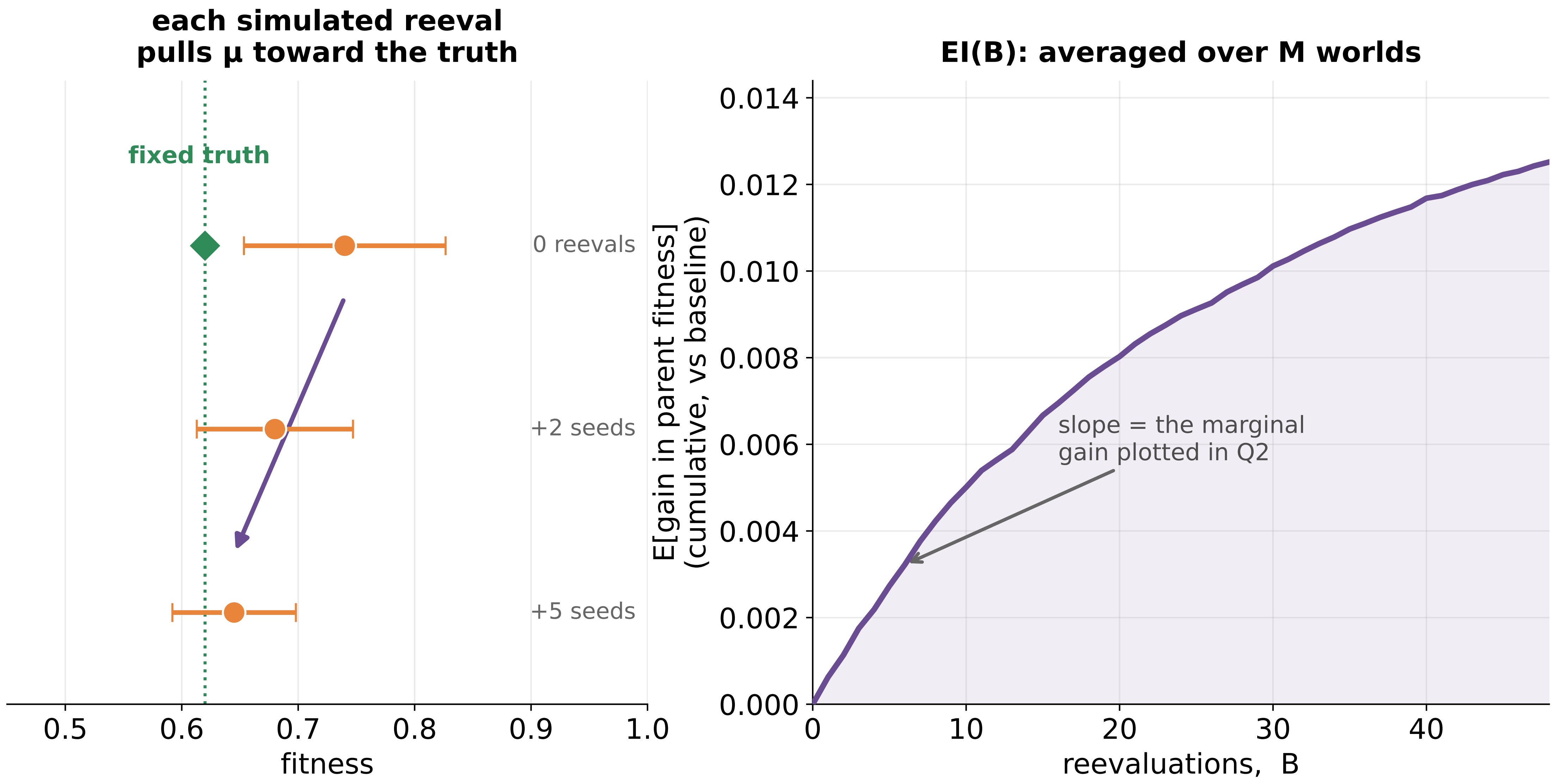
① Draw one plausible true value per arm (◆) = one simulated world.

② Baseline = true fitness of the parent today's estimates would pick.

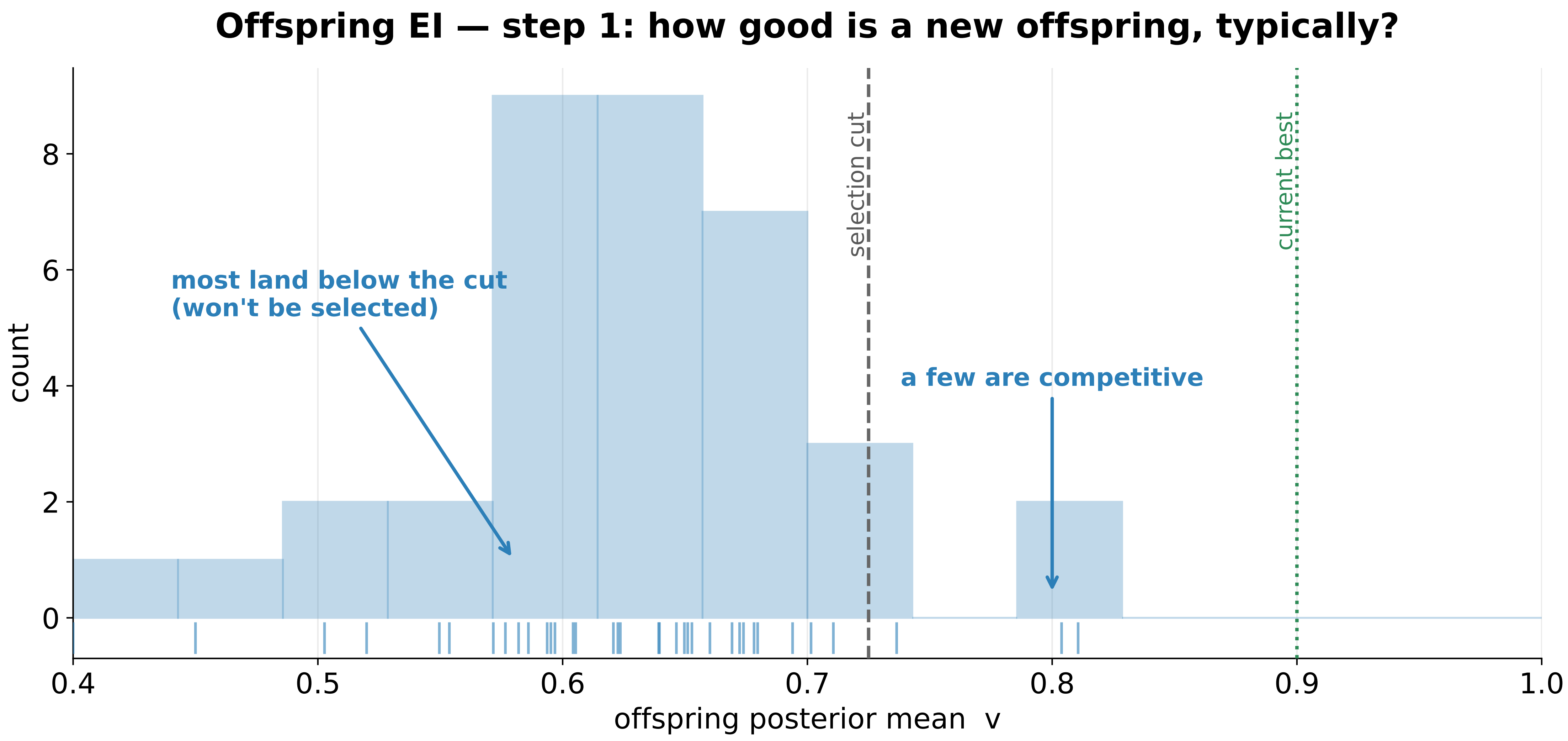
Repeat for $M \approx$ thousands of worlds.

The sampled truth is the hidden fitness in that imagined world; reevaluations (next slide) reveal noisy glimpses of it.

Reevaluation EI — step 2: simulate the extra seeds

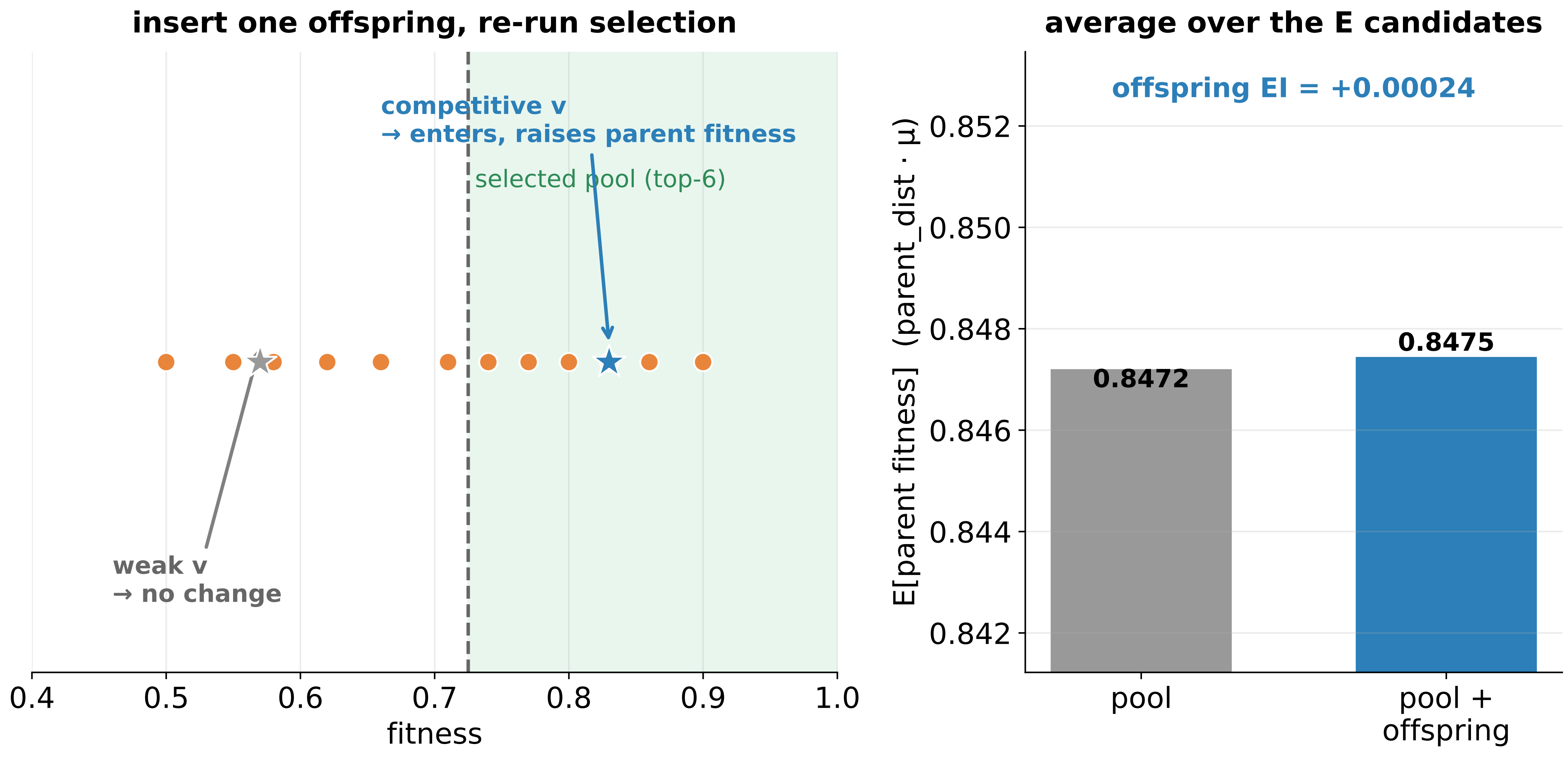


Per world: TTTS picks an arm → draw a seed ~ N(truth, σ) → update μ → re-pick the parent → score its truth vs the baseline.



We can't know a new offspring's value, so we sample from the empirical spread of recent offspring (last K=3 gens, E=36 values).

Offspring EI — step 2: insert each candidate, re-select



Analytic: parent_dist · μ averaged over the E candidates (no truth-sampling — the pool is fixed). This is Q2's offspring line.

Expected parent-fitness improvement — the math

Setup

Pool of k arms. Arm i: posterior $\theta_i \sim \mathcal{N}(\mu_i,\ \sigma^2/N_i)$ from N_i noisy seeds.

Parent-selection rule $\pi(\mu)$: probability of picking each arm (top-k + tournament).

Objective = true fitness of the selected parent: $\mathrm{PF}(\mu;\theta) = \sum_i \pi_i(\mu)\,\theta_i = \pi(\mu)\cdot\theta$

One evaluation, two uses — reevaluate an existing arm, or spend $\Delta = n_{init}$ seeds on a new offspring.

Reevaluate (Monte-Carlo)

$$\mathrm{EI}_{re}(B) = \mathrm{E}_\theta[\,\pi(\mu^{(B)})\cdot\theta\,] - \mathrm{E}_\theta[\,\pi(\mu^{(0)})\cdot\theta\,]$$

per world $\theta \sim \mathcal{N}(\mu, \sigma^2/N)$, then for $b = 1 \ldots B$:

pick $a \sim \psi_{TTTS}$, seed $y \sim \mathcal{N}(\theta_a, \sigma)$

$$\mu_a \leftarrow \frac{N_a\mu_a + y}{N_a + 1}, \quad N_a \leftarrow N_a + 1$$

marginal: $\mathrm{MEI}(B;\Delta) = \mathrm{EI}_{re}(B{+}\Delta) - \mathrm{EI}_{re}(B)$

Add one offspring (analytic)

draw $v \sim$ empirical offspring means $\{v_1, \ldots, v_E\}$

extend pool with arm $(\mu = v,\ N = n_{init})$

$$\mathrm{EI}_{off} = \frac{1}{E}\sum_e \pi(\mu^{+v_e})\cdot\mu^{+v_e} \ - \ \pi(\mu)\cdot\mu$$

pool fixed per candidate $\Rightarrow$

$\mathrm{E}[\pi\cdot\theta] = \pi\cdot\mu$ (no truth sampling)

Decision

Reevaluate while one reeval still beats one offspring: $B^* = \min\{\,B :\ \mathrm{MEI}(B;\Delta) \le \mathrm{EI}_{off}\,\}$